



Chapter 3: Hedging Strategies Using Futures - Comprehensive Summary

This chapter explored the core purpose of futures markets: to provide mechanisms for companies and investors to **reduce or offset specific price risks** (exposures) they face in their primary business activities.

I. Basic Hedging Principles

The goal of hedging is to take a position in the futures market whose gains/losses offset the losses/gains in the real world (spot market), locking in a desired price or cost.

A. Types of Hedges

Hedge Type	Futures Position	When Appropriate (Your Exposure)
Short Hedge	Sell futures contracts (Short Position)	When the company gains if the asset price increases and loses if the asset price decreases (e.g., you own the asset and plan to sell it later).
Long Hedge	Buy futures contracts (Long Position)	When the company loses if the asset price increases and gains if the asset price decreases (e.g., you need to buy the asset later).

II. Arguments for and Against Hedging

While risk reduction should be universally welcomed, companies often face practical and theoretical resistance to hedging.

A. Reasons Why Companies Hedge (Pro)

- Focus on Core Competency:** Allows management to concentrate on their primary business (manufacturing, service delivery) rather than speculating on financial or commodity price variables (e.g., interest rates, exchange rates).
- Avoid Surprises:** Stabilizes cash flow and profit margins by eliminating exposure to sharp, adverse price movements.

B. Reasons Why Companies May Not Hedge (Con)

- Shareholder Diversification (Theoretical):** Shareholders holding well-diversified portfolios have already eliminated much of the company's systematic risk. They may not require the company to hedge the same risk.
- Competitive Risk (Practical):** If a company hedges but its competitors do not, the hedged company's profit margin may actually **fluctuate more** than its competitors' margins, making it appear risky in the short term.
- Executive Criticism (Internal Risk):** If the price of the underlying asset moves favorably (i.e., the company makes a gain on the asset), the corresponding loss on the futures hedge may draw criticism from management or shareholders who do not understand the strategy's purpose.

III. Managing Basis Risk

An important reality is that perfect hedges are rare, primarily due to **Basis Risk**.

A. Definition of Basis

The basis is the difference between the asset's price in the real world and the price of the hedging contract:

$$\text{Basis}(b) = \text{Spot Price of Asset to be Hedged}(S) - \text{Futures Price of Contract Used}(F)$$

B. Basis Risk

Basis Risk is the uncertainty surrounding the **value of the basis** at the time the hedge is closed out.

- **Impact:** The final effective price received or paid is $F_{initial} + b_{final}$. Since $F_{initial}$ is known, the final basis (b_{final}) is the source of residual risk.
- **Causes:** The two prices (Spot and Futures) may not move perfectly together due to:
 - i. The hedging asset being different from the underlying asset (**Cross Hedging**).
 - ii. The futures contract being closed out before its delivery month.

IV. Optimal Hedging and Quantification

A. The Hedge Ratio

The **Hedge Ratio (h or h^*)** is the ratio of the size of the futures position to the size of the exposure ($\frac{\text{Size of Futures Position}}{\text{Size of Exposure}}$).

- If the assets are identical, the hedge ratio is often 1.0.
- **Minimum Variance Hedge Ratio (h^*):** When assets are different (**Cross Hedging**), the optimal h^* minimizes the variance (risk) of the final hedged position.
 - h^* is calculated as the **slope of the best-fit line** obtained by regressing daily changes in the spot price (ΔS) against daily changes in the futures price (ΔF).

$$h^* = \rho \frac{\sigma_S}{\sigma_F}$$

- **Optimal Number of Contracts (N):** Once h^* is found, the number of contracts is:

$$N = \frac{h^* \times \text{Exposure Quantity}(Q_A)}{\text{Futures Contract Size}(Q_F)}$$

V. Stock Index Futures for Equity Hedging

Stock index futures are essential for managing **systematic risk** (market risk) in stock portfolios.

A. Hedging with Beta (β)

The appropriate hedge size for an equity portfolio is determined by the portfolio's **Beta (β)**, which measures its sensitivity to the index.

- **Formula for Number of Contracts (N):**

$$N = \beta \frac{V_A}{V_F}$$

Where V_A is the portfolio value and V_F is the value of one futures contract (Futures Price $\times$ Multiplier).

B. Changing Portfolio Beta

Futures allow fund managers to adjust their market exposure **without buying or selling the underlying stocks** (which saves transaction costs):

Strategy Goal	Target Beta (β_T)	Action	Purpose
Complete Hedge	$\beta_T = 0$	Short $N = \beta \frac{V_A}{V_F}$ contracts	Removes all market risk; locks in return near the risk-free rate; isolates stock-specific performance (alpha).
Increase Risk	$\beta_T > \beta$	Long $N = (\beta_T - \beta) \frac{V_A}{V_F}$ contracts	Leverages the portfolio; provides immediate market exposure for a very bullish view.
Decrease Risk	$\beta_T < \beta$	Short $N = (\beta - \beta_T) \frac{V_A}{V_F}$ contracts	Reduces volatility for a moderate bearish or highly cautious outlook.

VI. Stack and Roll Strategy

When the required hedge period is longer than any liquid futures contract available, the hedge must be rolled forward.

- **Procedure:** Enter a sequence of short-dated futures contracts. As the near contract approaches expiration, **close it out** and **immediately enter a new contract** with a later delivery month.
- **Result:** This effectively creates a long-dated futures position by stacking a series of short ones.
- **Primary Risk: Liquidity Risk** (Cash Flow Mismatch). If prices move adversely (e.g., falling in a long hedge), the daily margin calls create immediate cash outflows that may be difficult to meet, even if the long-term hedge proves profitable (as illustrated by the **Metallgesellschaft disaster**).